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Learning multi-task local metrics for image annotation

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Abstract The goal of image annotation is to automatically assign a set of textual labels to an image to describe the visual contents thereof. Recently, with the rapid increase in the number of web images, nearest neighbor (NN) based methods have become more attractive and have shown exciting results for image annotation. One of the key challenges of these methods is to define an appropriate similarity measure between images for neighbor selection. Several distance metric learning (DML) algorithms derived from traditional image classification problems have been applied to annotation tasks. However, a fundamental limitation of applying DML to image annotation is that it learns a single global distance metric over the entire image collection and measures the distance between image pairs in the image-level. For multi-label annotation problems, it may be more reasonable to measure similarity of image pairs in the label-level. In this paper, we develop a novel label prediction scheme utilizing multiple label-specific local metrics for label-level similarity measure, and propose two different local metric learning methods in a multi-task learning (MTL) framework. Extensive experimental results on two challenging annotation datasets demonstrate that 1) utilizing multiple local distance metrics to learn label-level distances is superior to using a single global metric in label prediction, and 2) the proposed methods using the MTL framework to learn multiple local metrics simultaneously can model the commonalities of labels, thereby facilitating label prediction results to achieve state-of-the-art annotation performance.

Keywords Image annotation · Label prediction · Metric learning · Local metric · Multi-task learning

1 Introduction

In recent years we have witnessed an explosion in the number of images and videos resulting in a continuous increase in the huge volume of multimedia data in personal archives, social

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media and content-sharing websites. Automated methods are needed to handle this huge volume of data for retrieval, annotation and visualization purposes. In this paper we focus on the problem of image annotation, where the goal is to automatically assign a set of relevant labels from a given vocabulary to an image to describe the objects appear in the image. Image annotation is an important technique that exploits many practical applications, such as indexing, image retrieval and multimodal search.

Recently, nearest neighbor (NN) based methods such as [14, 20, 22, 31, 32, 40] have been successfully applied to image annotation problems and achieve inspiring results despite their simplicity. The intuition of the NN based methods is that "similar images share common labels" [22], and this similarity is usually measured by computing the distance between two images. To predict labels for an unlabeled test image, the NN based methods first select k nearest neighbors for the test image from the training set, and then transfer the labels from neighbor images to the test image according to the designed label prediction scheme.

Generally, the label prediction result of the NN based methods depends on the efficiency of neighbor selection. To select appropriate neighbors, several NN based methods such as [14, 22, 31, 32] utilize the distance metric learning (DML) algorithms, which are derived from classification problems, to find a global optimal metric for image-level similarity measure from a collection of labeled images. In addition, these methods adopt a weighted NN based label prediction scheme that takes a weighted average of labels' presence among the neighbors, where the weight of each label is determined by the similarity score between the corresponding neighbor and the test image. However, in multi-label annotation setting, it may not be sufficient to use a single global metric to measure label-level similarities, because highly nonlinear multi-class decision boundaries may not be modeled adequately by a single distance metric. Moreover, the weighted NN based label prediction scheme assigns weight to label based on the image-level similarities, which is not able to discriminate different labels in a neighbor.

To facilitate understanding, we present an intuitive example in left rectangle of Fig. 1 to demonstrate the framework of label prediction using a single global metric in these methods. The three images on the left are neighbor images selected using a global metric. We can see that in this case, labels in each neighbor image have identical predicted distance values as shown on the arrows. This means that each of the labels in a particular neighbor has an equal similarity score (weight) when predicting labels for the test image. Thus, when some related labels in one neighbor are assigned to the test image, the other unrelated labels in this neighbor may also be assigned inevitably. For example, the labels "*cliff*", "*house*", "*road*", and "*tree*" in green text in the second neighbor image could be secondary candidates for the test image since they have the same image-level distance as the label "*sea*" in blue text; however, these labels are semantically unrelated to the test image.

In order to discriminate different labels in each neighbor, it is more reasonable to measure the label-level distance using a label-specific local metric instead of a global metric. In the right rectangle of Fig. 1, we show the framework of label prediction using multiple local metrics. It can be observed that the distances between pairwise images are label-level and measured using the corresponding label's local metric. In other words, each label in a neighbor has a discriminative distance value that reflects the true label-level similarity based on the local metric for the corresponding label. Thus, in this condition, the label prediction result for the test image would be more accurate since the influences of unrelated labels are efficiently removed.

Learning multiple local metrics instead of a single global metric was initially proposed by Weinberger et al. [34] to improve k nearest neighbor classification. Xu et al. [37] further

extended it to image annotation task and proposed to learn multiple label-specific local metrics based on pre-defined subset of training set for each label. This method was shown to attain better results than those of single global metric methods on benchmarking datasets such as Corel5K and ESP Game. Despite inspirational improvement of performance, this method indeed has several limitations. First, in [37] each label-specific local metric is learnt independently, which implies that useful commonalities of labels are ignored. Imagine that an image assigned with the labels "*chair*", "*plate*", "*cup*" and "*table*" strongly suggests that these labels are semantically correlated, but these commonalities are not leveraged when learning appropriate local metric for each label. Second, an additional calibration step is adopted in [37] to calibrate the distances measured by different label-specific metrics in the same scale. However, this additional procedure of calibration may potentially result in information loss and the re-scaled distances may not reflect the true similarities under the semantics of specific labels.

To overcome the limitations of [37], in this paper, we present two local metrics learning methods in multi-task learning (MTL) framework for image annotation task. Specifically, in our proposed methods, learning a specific local metric for each label is treated as a learning task. To further leverage the commonalities of labels, multiple learning tasks for different labels are integrated in the MTL framework. The advantages of the proposed methods are three folds: (1) Multiple learning tasks are jointly learnt in the proposed methods, which ensures that the commonalities of labels are explicitly captured when formulating local metric for each label. (2) In the proposed methods, distances of pairwise images measured by learnt local metrics are in the same scale, which indicates that the additional procedure of calibration in [37] is not required any more. (3) The proposed methods can be considered

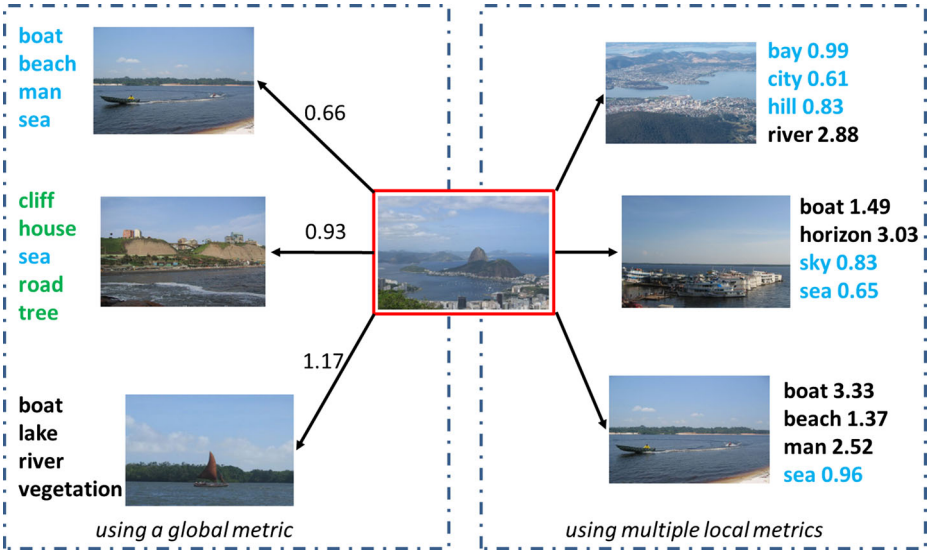


Fig. 1 Two frameworks of label prediction: using a global metric and multiple local metrics. The center image outlined in red is a test image, while the others are its neighbors selected using a global metric (three images on the *left*) and multiple local metrics (three images on the *right*). The labels in *blue text* are the final predicted labels for the test image, while labels in *green text* are secondary candidates. Numeric values represent the image-level or label-level distances. Note that a larger distance value indicates smaller similarity

as a unified solution to previous works involving global metric learning and independent local metric learning, since joint learning of multiple local metrics in the proposed methods ensures that both the uniqueness of each label and the generalities of multiple labels are incorporated in each local metric.

The main contributions of this works are summarized as follows: (1) We improve previous weighted NN based label prediction scheme using multiple label-specific local metrics. In our new scheme, each label in a neighbor has a more appropriate weight depending on its label-level similarity measured by corresponding local metric. (2) To learn multiple label-specific local metrics, we propose two multi-task local metrics (MTLM) learning methods using different learning manner: MTLM-LDML of discriminant manner and MTLM-LMNN of large margin manner. Both proposed methods have advantages in practice depending on the characteristics of dataset. (3) We further develop efficient optimization algorithms for the proposed methods.

The superiority of the proposed methods is verified on two challenging annotation datasets: ImageCLEF2011 [24] and NUS-WIDE [10]. Especially on the dataset ImageCLEF2011, which was used in the 2011 ImageCLEF photo annotation competition,[1] the proposed method achieves competitive annotation performance compared to published state-of-the-art approaches.

The rest of the paper is organized as follows. Section 2 gives an overview of previous works that are most related to ours. Section 3 first introduces the label prediction scheme using multiple label-specific local metrics, and then describes the proposed two local metric learning methods in the MTL framework. Section 4 presents the extensive experiments and comprehensive comparisons of proposed methods with baselines and state-of-the-art methods. Finally, we conclude the paper and discuss future research in Section 5.

2 Related work

In this section we review the related work on image annotation, distance metric learning (DML) and multi-task learning (MTL). Given the rich literature on the three subjects, we only discuss studies closely related to this work, and more detailed surveys of the three topics can be found in [6, 21, 22].

2.1 Image annotation

According to [13, 14, 20, 32], previous researches for image annotation can be roughly categorized into three groups: generative models, discriminative models and nearest neighbor (NN) based models. The generative models define the joint distribution over image features and labels with various probabilistic assumptions. In [4, 27], topic model such as latent Dirichlet allocation was proposed assuming that image features and annotation labels are associated in the latent topic space. Feng et al. [12] proposed a generative learning approach based on multiple Bernoulli relevance model. Notably in [36], Xiang et al. proposed a Markov Random Field based approach and showed that it can flexibly accommodate most of previous generative models. Generally, the generative models are sophisticated and may require a large quantity of training data and a time-consuming model training process to learn the joint probability of semantic concepts and image visual features.

[1] For more details see: http://www.imageclef.org/2011/Photo.

The discriminative models treat image annotation as a classification problem and learn a separate classifier for each label. Carneiro et al. [5] proposed a probabilistic formulation for image annotation by defining it as a multi-class multi-labelling problem with each label being a class. Verma et al. [33] explored the efficiency of traditional support vector machines for image annotation by exploiting the visual and semantic associations of image features and annotation labels. The disadvantage of discriminative models is that it is not easy to model the correlations between labels and one will encounter difficulties when extending to large number of labels.

On the other hand, the NN based models [14, 18, 20, 22, 32], which propagate labels to a test image from its visual neighbors, tend to be more attractive and preferable due to their concision and effectiveness with the rapidly-increasing image data. Among the NN based methods, Li et al. [18] proposed to search visual neighbors from web images and utilized a joint-modality label relevance estimation method to select most relevant labels for test image. Lin et al. [20] used a constraint range to allow more flexible choices of neighbors and proposed a label prediction method called TagSearcher which efficiently considers both textual and visual clues. Makadia et al. [22] proposed a greedy algorithm termed JEC that simply propagates labels to a test image with frequent label in its nearest visual neighbors. Guillaumin et al. [14] further presented a label prediction method named Tagprop that transfers labels to a test image by taking a weighted average of labels' presence among its neighbors. Tagprop also utilized metric learning for better weight distribution among visual neighbors and trained label-specific discriminant models to boost the annotation for infrequent labels. In our work, we propose two methods which also belong to the NN based models. These methods are efficiently designed to obtain more appropriate neighbors and predict more accurate labels for the test image by learning specific local metrics in the multi-task learning framework.

2.2 Distance metric learning for image annotation

For NN based models, neighbor selection is highly dependent on the metric used to compute the distance of pairwise images, thus learning an appropriate distance metric from labeled images is important. Here we highlight several methods exclusively for learning metrics for annotation tasks. On one hand, several algorithms have been developed to learn a linear distance metric from a collection of images. Makadia et al. [22] first used an average of the base distance metrics computed from multiple features (such as L_1 for color histogram, χ^2 for SIFT) for neighbor selection. Guillaumin et al. [14] designed a distance function to model the weight of each neighbor image, and developed a learning algorithm to obtain the optimal Mahalanobis distance metric. Verma et al. [32] proposed a new distance metric that linearly combines multiple base distance metrics and solved the metric learning problem using a large-margin nearest neighbor (LMNN) [34] algorithm. Mensink et al. [23] further extended LMNN algorithm to represent high dimensional features and learnt a low-rank Mahalanobis distance to reduce learning complexity. Xu et al. [37] used a logistic discriminant metric learning (LDML) [15] method to learn the linear combination of base distance metrics. On the other hand, a number of non-linear DML approaches have recently been proposed to handle high dimensional data. Wu et al. [35] carried out probabilistic DML to learn a non-linear distance metric by exploiting the side information of web images, while Feng et al. [13] summarized the kernel metric learning (KML) methods and proposed a regression based KML algorithm which is able to handle large-scale image data with high dimensional features.

As mentioned previously, a significant commonality of previous metric learning methods is that they learn a single global distance metric from the entire image collection. Indeed, to explore the discrimination of labels for multi-label annotation problems, it is necessary to learn more appropriate label-specific local metrics.

2.3 Multi-task learning for image annotation

The idea behind MTL is that, if the learning tasks are related in some sense, it may be advantageous to consider these relations among tasks and learn multiple tasks simultaneously. It has been empirically and theoretically demonstrated that learning multiple related tasks jointly usually result in better performance than learning each task independently [1, 2, 6]. When considering classifying each label as a task, many traditional classification algorithms could be extended to the MTL paradigm. For examples, Evgeniou et al. [11] improved the conventional support vector machine (SVM) for document classification problems using an MTL framework; Paraemeswaran et al. [26] embedded the LMNN algorithm in the MTL framework for KNN classification; Yuan et al. [39] explored multi-task joint sparse representation for visual classification problems in a multi-feature fusion setting while Yang et al. [38] proposed a new multi-task feature selection algorithm using shared knowledge from multiple types of features, and verified the efficiency of their approach on action and face recognition problems.

For image annotation problems, Binder et al. [3] combined SVM classifiers' outputs of labels and designed an effective approach, called output kernel multi-task learning, which achieved the best performance in ImageCLEF2011 photo annotation contest. Lin et al. [19] and Chen et al. [9] focused on annotating images from multiple image datasets by considering the learning model from each dataset as a separate task. They took advantage of the MTL framework to capture shared information in all the datasets, which can efficiently relieve the risky bias of a single dataset and improve annotation results. However, the efficiency of the MTL framework on local metric learning has not yet been explored for image annotation tasks, indeed it would be helpful to leverage the MTL framework since learning label-specific local metrics jointly can capture shared information explicitly and formulate more appropriate local metrics bridging the gap between the semantic and visual spaces.

3 Proposed methods

In this section we presents our proposed scheme for image annotation. We first give an overview of the proposed scheme, and then we describe the label prediction for test image using multiple label-specific local metrics in detail. Finally, we develop two local metrics learning algorithms under multi-task learning framework.

3.1 Overview of proposed scheme

Let $\{I_1, ..., I_N\}$ be a collection of N images and $\mathcal{Y} = \{y_1, ..., y_M\}$ be a vocabulary of M labels. The training set $\mathcal{S} = \{\{I_1, Y_1\}, ..., \{I_N, Y_N\}\}$ consists of images with their corresponding label sets, where each $Y_i \subseteq \mathcal{Y}$, $i \in \{1, ..., N\}$ contains multiple labels. Our goal is to learn an annotation model from training set $\mathcal{S}$, and then predict labels for an unlabeled test image J via learnt annotation model. Before discussing our proposed scheme of learning annotation model and predicting labels, we first define some important concepts.

Definition 1 (Semantic cluster) For label $y_m \in \mathcal{Y}$, $m \in \{1, ..., M\}$, its semantic cluster $\mathcal{S}_{y_m} \subseteq \mathcal{S}$ contains all images annotated with label y_m in $\mathcal{S}$. Note that different semantic clusters are not disjoint, as an image may have multiple labels and hence belongs to multiple semantic clusters.

Definition 2 (Semantic neighborhood) For a test image J, we first pick K_1 images from each semantic cluster that are nearest to J and form corresponding sets $\mathcal{N}_{J,y_m} \subseteq \mathcal{S}_{y_m}$, $m \in \{1, ..., M\}$. The images in each set $\mathcal{N}_{J,y_m}$ are the semantic neighbors of J corresponding to label y_m. Then the semantic neighborhood of J is defined as $\mathcal{N}_J = \{\mathcal{N}_{J,y_1} \bigcup ... \bigcup \mathcal{N}_{J,y_M}\}$.

Figure 2 illustrates our proposed scheme for image annotation, which consists of training stage and testing stage. In the training stage, at first, the training set $\mathcal{S}$ are partitioned into multiple label-specific semantic clusters $\mathcal{S}_{y_m}$, $m \in \{1, ..., M\}$, to ensure that images sharing a specific label are grouped into a local cluster. Then based on each label's semantic cluster, a local metric is learnt by considering it as a learning task in proposed multi-task local metrics learning methods. Note that the learning tasks of all labels are learnt jointly under the multi-task learning framework, to ensure that the commonalities of all labels are modeled explicitly. In Section 3.4, we proposed two multi-task local metrics (MTLM) learning methods based on different learning manners: MTLM-LDML of discriminant manner and MTLM-LMNN of large margin manner. Specifically, the module "metric learning" in Fig. 2 represents proposed MTLM-LDML and MTLM-LMNN methods, and different tasks can only use the same metric learning method simultaneously, either MTLM-LDML or MTLM-LMNN.

In the test stage, given a test image J, we formulate the semantic neighborhood $\mathcal{N}_J$ of J by selecting neighbor images from each label's semantic cluster of training set. And these neighbor images are obtained by using learnt label-specific local metrics. Then we predict labels for J based on $\mathcal{N}_J$ using a novel label prediction scheme which is described in Section 3.2. It is worth noting that our label prediction scheme leverages multiple label-specific local metrics for distance measure rather than a single global metric.

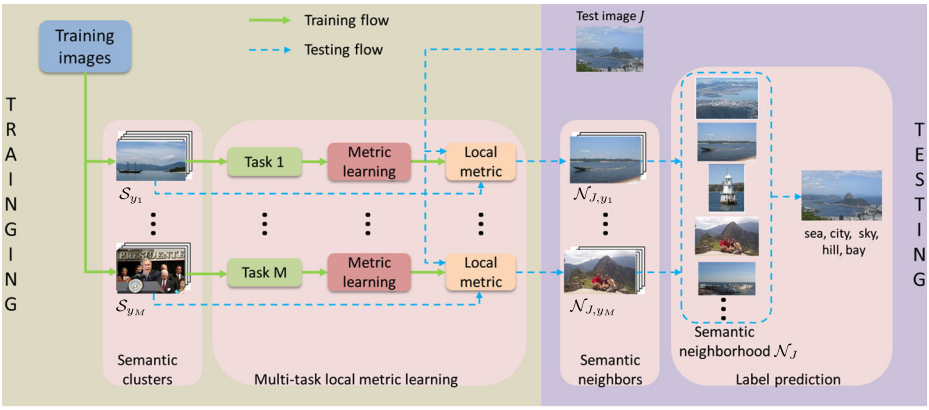


Fig. 2 Schematic illustration of the proposed scheme for image annotation. The *solid green line* represents training flow, and *dash blue line* represents testing flow. The training images are divided into multiple label-specific semantic clusters, and then a multi-task local metric learning framework is adopted to jointly learn local metric for each label based on corresponding semantic cluster. Finally, neighbor selection and label prediction processes are performed for the test image based on the learnt label-specific local metrics

3.2 Label prediction using label-specific local metrics

To annotate an unlabeled test image J, previous NN based works [14, 32] first retrieve K nearest neighbors from training set $\mathcal{S}$ as the neighborhood $\mathcal{N}_J = \{(I_1, Y_1), ..., (I_K, Y_K)\}$ of J, and use labels assigned to samples in $\mathcal{N}_J$ as candidates for label prediction. Then the presence/absence of each label in J is considered as the weighted average over K neighbors in $\mathcal{N}_J$. Here the weight of the k-th neighbor I_k to J is denoted as $\pi(J, I_k)$, where $\pi(J, I_k) \geq 0$ and $\sum_{k=1}^{K} \pi(J, I_k) = 1$. And $\pi(J, I_k)$ is defined as a smooth exponential function over the image-level distance $D(J, I_k)$, as

$$\pi(J, I_k) = \frac{\exp\left(-D(J, I_k)\right)}{\sum_{k'=1}^{K} \exp\left(-D(J, I_{k'})\right)}. \tag{1}$$

Thus, for each label y_m, $m \in \{1, ..., M\}$, its presence probability in J is formulated as,

$$\begin{aligned} P(y_m = +1|J) &= \sum_{k=1}^{K} \pi(J, I_k) \cdot \delta(y_m \in Y_k|I_k) \\ &\propto \sum_{k=1}^{K} \exp\left(-D(J, I_k)\right) \cdot \delta(y_m \in Y_k|I_k), \end{aligned} \tag{2}$$

where Y_k is the corresponding label set of I_k, $\delta(y_m|I_k)$ is an indicator function which denotes the presence/absence of label y_m in I_k, with $\delta(\cdot) = 1$ when $y_m \in Y_k$ and $\delta(\cdot) = 0$ otherwise. By ranking the probabilities of all labels $\{y_m\}_{m=1}^{M}$ based on (2), top ranked labels can be assigned to J.

It can be learnt from (2) that, in [14, 32] multiple labels Y_k in I_k have the same weight $\exp\left(-D(J, I_k)\right)$ since the image-level distance $D(J, I_k)$ is uniquely measured by the global distance metric $D(\cdot)$. Thus the importance of different labels in Y_k can not be distinguished. This may lead the label prediction result to be sensitive to noisy labels in the neighbors, as the previous illustration in Fig. 1. To overcome the limitation of the single global metric used in previous works [14, 32], we are motivated to exploit the more appropriate label-level distance $D_{y_m}(J, I_k)$ for specific label $y_m \in Y_k$ in I_k, in which $D_{y_m}(J, I_k)$ is measured by the specific local distance metric $D_{y_m}(\cdot)$ of label y_m. Different from previous works [14, 32], we first generate local neighborhood $\mathcal{N}_J = \{(I_1, Y_1), ..., (I_{K^*}, Y_{K^*})\}$ using label-specific local metrics $D_{y_m}(\cdot)$, $m \in \{1, ..., M\}$, where $K^* = K_1 \times M$ denotes the number of images in $\mathcal{N}_J$ (see Definition 2 in Section 3.1). Then the weight of label y_m in I_k is given by

$$\pi_{y_m}(J, I_k) = \frac{\exp\left(-D_{y_m}(J, I_k)\right)}{\sum_{k'=1}^{K^*} \exp\left(-D_{y_m}(J, I_{k'})\right)}. \tag{3}$$

Thus, based on the label-dependent weight of each label for each neighbor in (3), we can predict the presence of m-th label y_m in J considering its probability as follows:

$$P(y_m = +1|J) = \sum_{k=1}^{K^*} \exp\left(-D_{y_m}(J, I_k)\right) \cdot \delta(y_m \in Y_k|I_k). \tag{4}$$

Differing from previous prediction rule of (2), the weight $\exp\left(-D_{y_m}(J, I_k)\right)$ depends on the specific label y_m. In particular, we can measure the label-level distances between neighbor I_k and J using the local metrics of labels $\forall y_m \in Y_k$ in I_k. This allows us to explore the discriminations of different labels, based on the semantic similarity of images I_k and J.

3.3 Label-specific distance metrics in multi-task framework

In the previous work [37], label-specific local metrics are defined as linear combinations of base distance measures over multiple features (e.g., L_1 for color histogram, χ^2 for SIFT features). In this work, besides learning these label-specific local metrics, we also introduce an additional shared metric for all labels to model the commonalities of labels.

Specifically, for two images A, B represented by n types of features, where the dimension of i-th feature is N_i, $i = \{1, ..., n\}$, then the distance $D_{y_m}(A, B)$ between A and B under label y_m is considered as a combined distance measured by the local metric of y_m and the shared metric, calculated as

$$\begin{aligned} D_{y_m}(A, B) &= (\boldsymbol{w}_m + \boldsymbol{w}_*)\boldsymbol{d}(A, B) \\ &= \sum_{i=1}^{n} (\boldsymbol{u}_m^i + \boldsymbol{u}_*^i) \sum_{j=1}^{N_i} (\boldsymbol{v}_m^j + \boldsymbol{v}_*^j)\boldsymbol{d}^{i,j}(A, B). \end{aligned} \tag{5}$$

Here $\boldsymbol{w}_m$ is the weight vector of the local metric of y_m, $\boldsymbol{w}_*$ is the weight vector of the shared metric, and $\boldsymbol{d}(\cdot)$ is the base distance measure (e.g., L_1, L_2 and χ^2) written as a dot product of two feature vectors. Moreover, both $\boldsymbol{w}_m$ and $\boldsymbol{w}_*$ can be further decomposed into non-negative real-valued inter-feature weights $\{\boldsymbol{u}_m, \boldsymbol{u}_*\}$ and intra-feature weights $\{\boldsymbol{v}_m, \boldsymbol{v}_*\}$, and we add additional constraints to $\boldsymbol{u}_m$ and $\boldsymbol{v}_m$ that $\sum_{i=1}^{n} \boldsymbol{u}_m^i = n$ and $\sum_{j=1}^{N_i} \boldsymbol{v}_m^j = 1$, which ensure that the weights are normalized and never arbitrarily scaled.

It is worth noting that the definition of the local metric in (5) has several advantages: (1) The shared metric is able to model the commonalities of labels explicitly. (2) The combined form in (5) can be treated as a unified representation of both the local metric of y_m (label-specific) and the shared metric (global), thus the distance measure is more appropriate to the semantics of corresponding labels. (3) It is feasible to leverage the MTL framework to learn label-specific local metrics and shared metric jointly, whereas in previous work [37], the label-specific local metrics are learnt separately.

In the MTL framework, we consider learning local metrics $\{D_{y_m}(\cdot)\}_{m=1}^{M}$ of M labels as M tasks, and introduce an additional task to model the commonalities of M labels, which leads to learning the an additional shared metric for all M labels. The objective is to learn the non-negative weight vectors $\{\boldsymbol{w}_m\}_{m=1}^{M}$ of label-specific local metrics and $\boldsymbol{w}_*$ of the shared metric. To realize a trade-off between learning the label-specific local metrics and shared local metric, we use the regularization term for weights $\{\boldsymbol{w}_m\}_{m=1}^{M}$ and $\boldsymbol{w}_*$ similar to that in [26]:

$$\min_{\boldsymbol{w}_*, \boldsymbol{w}_1, ..., \boldsymbol{w}_M} \gamma_* \|\boldsymbol{w}_*\|_2^2 + \sum_{m=1}^{M} \gamma_m \|\boldsymbol{w}_m\|_2^2. \tag{6}$$

The trade-off parameters γ_* and $\{\gamma_m\}_{m=1}^{M}$ control the regularization of the weight vectors for all tasks. From (6) we see that if $\{\gamma_m\}_{m=1}^{M} \to \infty$ and γ_* is small, the learning of weight vector $\boldsymbol{w}_*$ of the shared metric is degraded across all tasks. Under this condition, (6) can be considered a single global metric learning method with $\boldsymbol{w}_*$ as the global metric. If $\gamma_* \to \infty$ and $\{\gamma_m\}_{m=1}^{M}$ are small, (6) reduces to learning the weight vectors $\{\boldsymbol{w}_m\}_{m=1}^{M}$ of M irrelevant label-specific metrics.

3.4 Learning multiple label-specific local metrics

3.4.1 Generating pairwise constraints locally

In distance metric learning, the training samples are cast into pairwise constraints: the equivalent constraints of similar pairs and the inequivalent constraints of dissimilar pairs [21]. The pairwise constraints are important since they determine how we measure the (dis)similarity for pairwise samples. Different from previous works [14, 22, 23, 32], which obtain pairwise constraints from the entire training set to learn a global metric, in this work we locally formulate pairwise constraints of each label from each label's semantic cluster, to learn label-specific local metrics.

Given the semantic cluster $\mathcal{S}_{y_m}$ of label y_m, for a labeled image (I_q, Y_q) in $\mathcal{S}_{y_m}$, we first generate its semantic neighborhood $\mathcal{N}_{I_q} = \{\mathcal{N}_{I_q,y_1} \bigcup ... \bigcup \mathcal{N}_{I_q,y_M}\}$ as described in Definition 2 in Section 3.1. And we define the similar neighbors of I_q as $\mathcal{N}^+_{I_q} = \bigcup \mathcal{N}_{I_q,y_q}, \forall y_q \in Y_q$, and the dissimilar neighbors of I_q as $\mathcal{N}^-_{I_q} = \bigcup \mathcal{N}_{I_q,y_q}, \forall y_q \in \mathcal{Y} \backslash Y_q$. Then the similar pairs of I_q is $\mathcal{P}^+_{I_q} = \bigcup\{(I_q, I_p)\}, \forall I_p \in \mathcal{N}^+_{I_q}$, and the dissimilar pairs of I_n is $\mathcal{P}^-_{I_q} = \bigcup\{(I_q, I_n)\}, \forall I_n \in \mathcal{N}^-_{I_q}$. Then the pairwise constraints of label y_m is defined as:

Definition 3 (Pairwise constraint) For label $y_m \in \mathcal{Y}$, $m \in \{1, ..., M\}$, its pairwise constraints $\mathcal{P}_{y_m}$ consist of similar pairs of images $\mathcal{P}^+_{y_m}$ and dissimilar pairs of images $\mathcal{P}^-_{y_m}$, where $\mathcal{P}^+_{y_m} = \bigcup \mathcal{P}^+_{I_q}$ and $\mathcal{P}^-_{y_m} = \bigcup \mathcal{P}^-_{I_q}, \forall I_q \in \mathcal{S}_{y_m}$.

Then metric learning is performed on the pairwise constraints of each label to learn corresponding local metric. Regarding the manner in which pairwise constraints are modeled for metric learning, previous works [15, 23, 32, 37] can be categorized as two types: discriminant manner and large-margin manner. For the discriminant manner (e.g. [15, 37]), the (dis)similarity measure is cast into a binary classification problem and a binary classifier is trained to discriminate similar/dissimilar pairs in the distance space. For large-margin manner (e.g. [23, 32]), the (dis)similarity measure is considered to find the margin in the distance space to pull similar pairs closer than dissimilar pairs. In our work, we adopt both manners to develop two label-specific local metric learning methods under the MTL framework.

3.4.2 Learning multiple local metrics in discriminant manner

In [15], Guillaumin et al. cast the global metric learning into a logistic discriminant problem given pairwise constraints, and the (dis)similarity measure is evaluated by posterior class probabilities. This method is called logistic discriminant metric learning (LDML). Following the discriminant manner in LDML, we further develop a new method under the multi-task local metric learning framework, called MTLM-LDML. The proposed MTLM-LDML learns one local metric for each label locally given the pairwise constraints of each label.

In particular, given an image I_q with its semantic neighborhood $\mathcal{N}_{I_q}$, we model the probability that I_q and sample $I_p \in \mathcal{N}_{I_q}$ are similar $s_{qp} = 1|y_m$ (or dissimilar, $s_{qp} = 0|y_m$) using the metric of label y_m, as:

$$p_{qp}(s_{qp} = 1|y_m; D_{y_m}(I_q, I_p), b_m) = \sigma(b_m - D_{y_m}(I_q, I_p)), \tag{7}$$

where $\sigma(z) = (1 + \exp(-z))^{-1}$ is the sigmoid function and b_m is a bias term.

Intuitively, given the pairwise constraints $\mathcal{P}_{y_m}$ of label y_m, the weight vector $\boldsymbol{w}_m$ of local metric $D_{y_m}(\cdot)$ can be obtained by optimizing the conditional log-likelihood as

$$\begin{aligned}\boldsymbol{w}_m &= argmax_{\boldsymbol{w}_m} \log \prod_{(I_q, I_p)\in\mathcal{P}_{y_m}} p_{qp}(s_{qp}|y_m; D_{y_m}(I_q, I_p), b_m) \\ &= argmin_{\boldsymbol{w}_m} \log \sum_{(I_q, I_p)\in\mathcal{P}_{y_m}} (1+\exp(-b_m + D_{y_m}(I_q, I_p))). \end{aligned} \tag{8}$$

Then, in the proposed MTLM-LDML method, we combine the regularizer in (6) with the objective in (8) applied to each of the M labels. Finally, the objective of MTLM-LDML is to minimize the following cost function:

$$\min_{\boldsymbol{w}_*, \boldsymbol{w}_1, \ldots, \boldsymbol{w}_M} \gamma_* \|\boldsymbol{w}_*\|_2^2 + m = 1 \sum^{M} \left[\gamma_m \|\boldsymbol{w}_m\|_2^2 + \sum_{(q,p)\in\mathcal{P}_{y_m}} \log(1+\exp(-s_{qp}(b_m - D_{y_m}(I_p, I_q)))) \right]. \tag{9}$$

It can be seen from (9) that, learning weight vectors $\{\boldsymbol{w}_m\}_{m=1}^{M}$ in each task has a similar optimization term as logistic regression with a non-negative constraint. Thus, the optimization problem in (9) can be efficiently solved using gradient descent based algorithm with projection steps. In Appendix A, we give a detailed description of the learning algorithm for the MTLM-LDML method.

3.4.3 Learning multiple local metrics in large margin manner

In [32], Verma et al. leveraged the large margin nearest neighbor (LMNN) method in image annotation task to learn a global distance metric in a large margin manner. Given a labeled image I_q and its similar neighbors $\forall I_p \in \mathcal{N}_{I_q}^{+}$ and dissimilar neighbors $\forall I_n \in \mathcal{N}_{I_q}^{-}$, the objective of LMNN is to find the optimal global distance metric that pulls I_p closer while pushing I_n further from I_q with a large margin.

Here we further extend LMNN to learn multiple local metrics under the multi-task local metric learning framework, and we propose a new model called MTLM-LMNN. The proposed MTLM-LMNN finds each label's local metrics to separate the targets and imposters of each sample locally with a large margin.

Specifically, given an image I_q with its similar pairs $\mathcal{P}_{I_q}^{+}$ and dissimilar pairs $\mathcal{P}_{I_q}^{-}$ in the pairwise constraints of the m-th label y_m, we want the distance of any similar pairs (I_q, I_p), $\forall I_p \in \mathcal{P}_{I_q}^{+}$, to be minimized and less than the distance of any dissimilar pairs (I_q, I_n), $\forall I_n \in \mathcal{P}_{I_q}^{-}$. Then the loss function for image I_q under the local metric of y_m is given by

$$E_{y_m}(I_q) = \sum_{(I_q, I_p)\in\mathcal{P}_{I_q}^{+}} D_{y_m}(I_q, I_p) + \mu \sum_{(I_q, I_n)\in\mathcal{P}_{I_q}^{-}} \left[1 + D_{y_m}(I_q, I_p) - D_{y_m}(I_q, I_n)\right]_+. \tag{10}$$

Here the term $[z]_+ = max(z, 0)$ denotes the standard hinge loss which punishes a dissimilar pair (I_q, I_n) with a small distance, while $\mu > 0$ handles the trade-off between the two terms. A margin of size 1 is used to ensure that the target I_p is closer than an imposter I_n.

Thus, by accumulating the loss in (10) of all images in the semantic cluster $\mathcal{S}_{y_m}$ of label y_m, the loss for label y_m can be formulated based on its pairwise constraints $\mathcal{P}_{y_m}$, as

$$E_{y_m} = \sum_{(I_q, I_p) \in \mathcal{P}^+_{y_m}} D_{y_m}(I_q, I_p) + \mu \sum_{(I_q, I_n) \in \mathcal{P}^-_{y_m}} \left[1 + D_{y_m}(I_q, I_p) - D_{y_m}(I_q, I_n)\right]_+. \quad (11)$$

Then, in the proposed MTLM-LMNN method, we combine the regularizer in (6) with the loss in (11) applied to each of the M labels. Finally, the objective of MTLM-LMNN is to solve the following constrained minimization problem:

$$\min_{\boldsymbol{w}_*, \boldsymbol{w}_1, \ldots, \boldsymbol{w}_M} \quad \gamma_* \|\boldsymbol{w}_*\|_2^2 + \sum_{m=1}^{M} \left[\gamma_m \|\boldsymbol{w}_m\|_2^2 + \sum_{(I_q, I_p) \in \mathcal{P}^+_{y_m}} D_{y_m}(I_q, I_p) + \mu \sum_{(I_q, I_n) \in \mathcal{P}^-_{y_m}} \xi_{qpn} \right] \quad (12)$$

$$\begin{aligned} subject\ to: \quad & D_{y_m}(I_q, I_n) - D_{y_m}(I_q, I_p) \geq 1 - \xi_{qpn}, \\ & \xi_{qpn} \geq 0, \quad \boldsymbol{w}_*, \boldsymbol{w}_1, \ldots, \boldsymbol{w}_M \succeq 0. \end{aligned}$$

Here $\succeq$ denotes entry-wise $\geq$, while ξ_{qpn} is a slack variable representing the hinge loss in (10). As the convexity of (11) has been verified in [34], and the three constraints in (12) are linear and positive defined, the optimization problem in (12) is convex. We refer to [32] and solve it using a sub-gradient descent algorithm with projection steps.

It should be noted that since calculating ξ_{qpn} involves triplets (q, p, n) from $\mathcal{P}_{y_m}$, and the set $\mathcal{P}_{y_m}$ is potentially very large, computing the loss and gradient of all triplets directly would be extremely expensive. Thus, we developed a scheme to update the loss and gradients relying only on "*active*" triplets, which are a small fraction of $\mathcal{P}_{y_m}$ changing from one iteration to the next. Details of the scheme, which is efficient and reduces the complexity in practice, are given in Appendix B.

4 Experiments

To demonstrate the effectiveness of the proposed multi-task local metric learning methods for image annotation in real world scenarios, we perform extensive experiments on two publicly available and challenging datasets ImageCLEF-11 and NUS-WIDE. Both datasets were collected from the Flickr website. In the following subsections, we first describe the experimental configurations of the datasets and the multiple features. Thereafter, we present the baselines, evaluation measures and implementation details. We further compare our proposed methods with baselines from various aspects.

4.1 Datasets and features

The ImageCLEF-11 dataset was used for the ImageCLEF2011 photo annotation challenge. The challenge requires 10,000 test images to be annotated according to 99 pre-defined semantic concepts, given a training set of 8,000 fully annotated images. The NUS-WIDE dataset is a large scale dataset comprising a total of 269,648 images with a ground-truth of 81 labels. Since the number of images is quite large, only original URLs of each image are provided. When downloading the raw images from Flickr, we found that some of the URLs were out of date, and finally obtained a subset of total 231,047 images (138,563 images for training and 92,484 images for testing). Table 1 summarizes the general statistics of the two cropped datasets we utilized in our experiments.

Table 1 General statistics for the two datasets used in this paper. The items in the second row are listed in the format "training/test", while items in the third and fourth rows are given in the format "mean/minimum/maximum"

	ImageCLEF-11	NUS-WIDE
Total labels	99	81
Number of images	8,000/10,000	138,563/92,484
Labels per image	11.51/3/26	1.76/1/20
Images per label	2,093/18/16,880	2,512/333/16,425

These two datasets are challenging in three folds: (1) Images are real world photographs with varying lighting conditions, quality and appearance. (2) The semantics of labels are heterogeneous and both datasets contain concrete concepts (e.g., "*cat*", "*car*") and abstract concepts (e.g., "*Boring*", "*Technical*"). (3) Some labels are even highly subjective, (e.g., "*Scary*", "*Still_Life*" in ImageCLEF-11), which are more complex for model learning than other objective concepts.

Since using multiple types of features including both global and local features can facilitate annotation performance, following related works [3, 28, 29], we also extract multiple features for each image in both datasets. Before computing the features, each image is resized to be at most 300×300 pixels, while maintaining the original aspect ratios. For global features, we extract Gist [25] features. For local features, we extract five kinds of visual descriptors, SIFT and color-SIFTs (C-SIFT, Opponent-SIFT, RGB-SIFT, RG-SIFT) using the public colorDescriptor tools [17]. The SIFT based features are computed without orientation invariance and the grid has a step size of three. The codebook for each SIFT based feature is generated from 7,000 randomly selected images, and quantized to 4,000 corresponding k-means clusters. Finally, concatenating the Gist features and bag-of-words (BoW) histograms of the local features, each image has a 20,512 dimensional feature vector. When measuring the base distance of features, we use the L_1 norm for the color-SIFT features (C-SIFT, Opponent-SIFT, RGB-SIFT, RG-SIFT), χ^2 norm for SIFT and L_2 norm for the Gist features.

4.2 Baselines and evaluation measures

For comparison with proposed methods, we consider seven baselines in our comparative study. These baselines include four global metric (GM) learning based methods, two label-specific local metric (LSLM) learning based methods and one SVM based method. In particular, the four GM learning based methods are: (1) JEC [22], which uses equal combination of multiple base distance jointly to select the nearest neighbors; (2) Tagprop [14], which learns the optimal linear combination of feature distances; (3) GM-LDML [15], which learns a global metric using the LDML algorithm in discriminant manner; and (4) GM-LMNN [32], which learns a global metric using the LMNN method in a large margin manner. The two LSLM learning based methods are: (1) LSLM-LDML [37], which is our previous work and learns label-specific local metrics independently in a discriminant manner; and (2) LSLM-LMNN, which learns label-specific local metrics independently in a large-margin manner. For the SVM based method, the one-versus-all training scheme is adopted to learn an SVM classifier for each label.

For the implementation of the baselines, we use the published code for baselines JEC, Tagprop and GM-LDML and implement GM-LMNN, LSLM-LDML, and LSLM-LMNN

referring to the instructions in the related papers. For the SVM based method, we use LIBSVM [7] to learn SVM classifiers with the chi-square kernel. The value of the SVM parameter C and the normalization factor of the chi-square kernel were determined by cross-validation. Regarding the initial neighborhood selection in the baselines and proposed methods, we use an equally weighted sum of base distances (normalized to have a maximum value of 1) of multiple features for the initial global metric (of JEC, Tagprop, GM-LDML, and GM-LMNN methods) and each local metric (of LSLM-LDML, LSLM-LMNN, MTLM-LDML, and MTLM-LMNN methods).

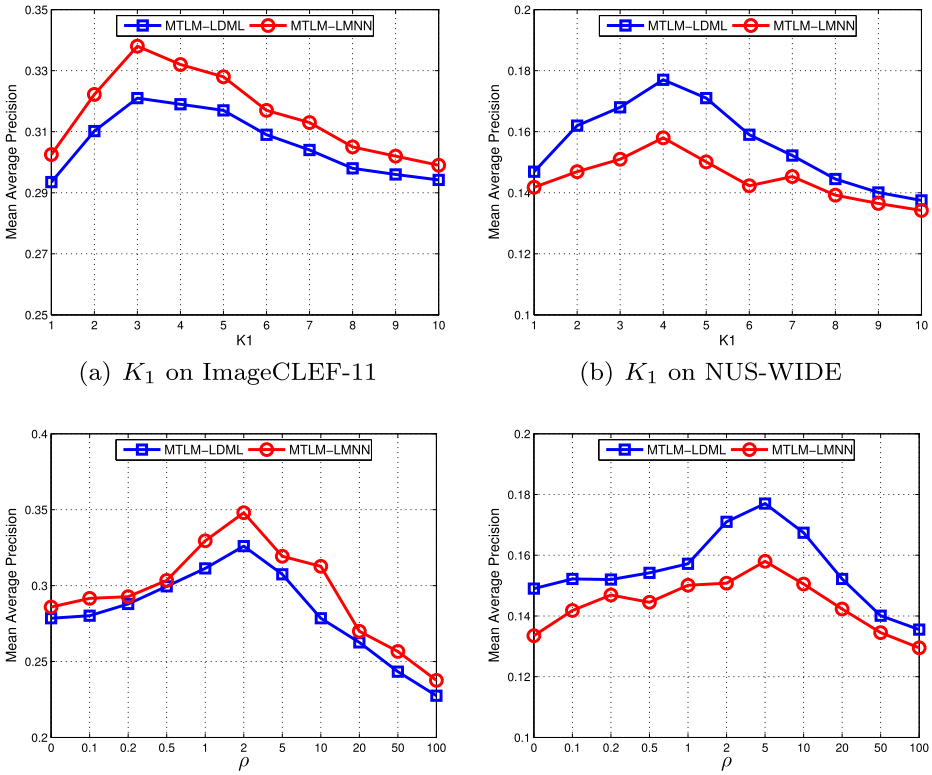
In both datasets, we take 20% of the original training data as the validation set, and conduct cross validation to tune the parameters. Measures for comparing the performance include the average precision (AP) of each label and mean average precision (MAP) of all labels. The former measure evaluates the annotations for each specific label, while the latter examines the overall trade-off performance for all labels. A larger numerical value for each measure indicates better annotation performance. All experiments are executed on a desktop PC equipped with a 3.4GHz CPU and 32GB physical memory.

4.3 Parameter settings

We first investigate the influence of parameter K_1, which is the number of nearest images selected from each label's semantic cluster when formulating an image's semantic neighborhood (see Definition 2 in Section 3.4.1). Since K_1 determines the size of the semantic neighborhood of the test image, it is an important factor affecting label prediction result. Specifically, for the proposed two methods, we initialize μ in MTLM-LMNN as 1 and $\gamma_*, \gamma_0, ..., \gamma_M$ as 1, and then learn the proposed models by varying K_1 from 1 to 10. Figure 3 a and b illustrates the influence of varying K_1 values on the two datasets. It can be seen that the optimal K_1 values for the two datasets are around 3 and 4, respectively. In addition, if K_1 is too large, more noise and irrelevant samples are selected in the semantic neighborhood which harms the prediction result.

Furthermore, we also assess the affect of the trade-off between the label-specific local metrics and the shared metric in (6), and we define the trade-off factor as $\rho = \frac{\frac{1}{M}\sum_{m=1}^{M}\gamma_m}{\gamma_*}$. Here we set K_1 to 3 and 4 for the two datasets, respectively, μ to 1 as in the previous setting, and ρ successively to the values $\{0, 0.1, 0.2, 0.5, 1, 2, 5, 10, 20, 50, 100\}$. As shown in Fig. 3c and d, the optimal settings for ρ are around 2 and 5 respectively for two datasets. Moreover, the shared metric in the proposed methods has the advantage of incorporating correlations of labels for label-specific distance measures. With ρ set to 0 and 100 the results of the MAP are inferior since in these cases, the proposed methods are degraded to learn M local metrics independently ($\rho = 0$) or to learn a unified global metric ($\rho = 100$).

It is interesting to see that the performance of the proposed two methods reverses on the two datasets. That is, on ImageCLEF-11, MTLM-LMNN performs better than MTLM-LDML for various values of K_1 and ρ, whereas on NUS-WIDE, the opposite is true. We attribute this to the different local metric learning manner used in the proposed two methods: the discriminant manner of MTLM-LDML and the large-margin manner of MTLM-LMNN. We further analyze the characteristics of these two learning manners in the following subsection.



(a) K_1 on ImageCLEF-11 (b) K_1 on NUS-WIDE

(c) ρ on ImageCLEF-11 (d) ρ on NUS-WIDE

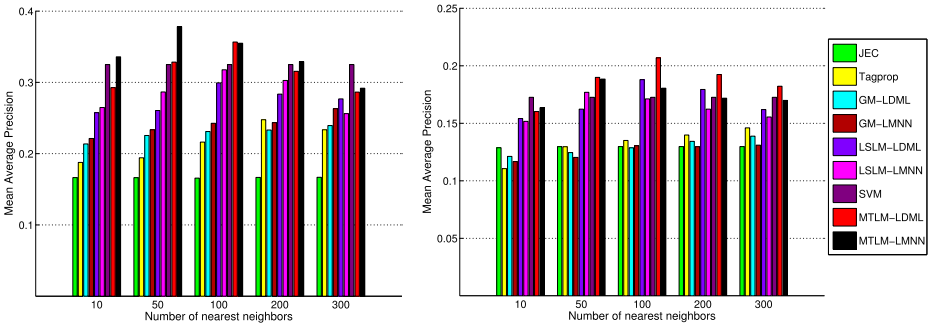
Fig. 3 Influence of parameters K_1 (**a**, **b**) and ρ (**c**, **d**) on the MAP for two datasets

4.4 Label prediction using multiple local metrics *vs.* single global metric

4.4.1 Effects of neighborhood selection

As the proposed methods and all baselines except for the SVM are KNN based methods, we first compare the label prediction results of the proposed methods and baselines by varying the number K of nearest neighbors used for prediction. For those methods using a single global metric (JEC, Tagprop, GM-LDML, and GM-LMNN), the K nearest neighbors are selected from the entire training data. For methods using multiple local metrics (LSLM-LDML, LSLM-LMNN, MTLM-LDML, and MTLM-LMNN), the K^* nearest neighbors are chosen from each label's semantic cluster (see the definition of K^* in Section 3.2). For example, on ImageCLEF-11, we first set $K_1 = 3$, which results in $K^* = 3 \times 99 = 297$ semantic neighbors of a test image, and then we select the top K nearest samples from the K^* semantic neighbors to perform label prediction.

Figure 4 shows the MAP of the two proposed methods and seven baselines with various values of K used for label prediction. And the following observations can be made: (1)



(a) ImageCLEF-11 (b) NUS-WIDE

Fig. 4 Comparison of MAP of the proposed methods and baselines varying the number K of nearest neighbors on the two datasets: ImageCLEF-11 (*left*), and NUS-WIDE (*right*). Setting $K = 300$ represents an approximate value of the maximum number of neighbor images used. Specifically, on ImageCLEF-11 we used all 297 neighbors, and on NUS-WIDE we used all 324 neighbors

Compared with baselines (JEC, Tagprop, GM-LDML and GM-LMNN) that learn a single global, the label prediction results of methods (LSLM-LDML, LSLM-LMNN, MTLM-LDML and MTLM-LMNN) that learn multiple local metrics improved significantly with respect to MAP. (2) When using multiple local metrics, the optimal number of nearest neighbors K_{opt} used to achieve the best MAP is remarkably reduced, compared with utilizing a global metric. More specifically, to achieve the best MAP on ImageCLEF-11, the values of K_{opt} for baselines (Tagprop, GM-LDML, and GM-LMNN) using a global metric are around 200, 300, and 300 respectively. On the contrary, for the proposed methods using multiple local metrics, MTLM-LDML requires about 100 neighbors, while MTLM-LMNN needs about 50 neighbors. Similar results are observed on the NUS-WIDE dataset. (3) For the methods using multiple local metrics, our proposed methods (MTLM-LDML and MTLM-LMNN), which embed learning label-specific local metrics in an MTL framework, consistently show improved performance compared with baselines LSLM-LDML and LSLM-LMNN.

We further analyze the differences in neighborhood selection when using multiple local metrics and a single global metric. Differences were identified in two respects: computing the distances of image pairs when selecting neighbors and label prediction using the distances in (4). When using a global metric, the model is limited to image-image similarity at the image level, and thus it tends to choose images that are visually similar but semantically irrelevant when formulating the neighborhood of the test image. On the other hand, using label-specific local metrics we are able to model image-label similarity as the distance is measured at the label level, and thus a more representative neighborhood is formulated. Figure 5 illustrates two typical test images of the two datasets and their ten nearest neighbors selected by a single global metric and multiple local metrics. It can be seen in Fig. 5 that neighbor images selected from local metrics are more semantically related to the two test images. In addition, these related neighbors obtain higher rank (smaller distance) in the neighborhood, whereas several irrelevant images get high rank in the neighborhood using a global metric.

A potential advantage of the proposed methods can be attributed to the integrated learning of all labels in the MTL framework. Unlike LSLM-LDML and LSLM-LMNN, which learn each label-specific local metric independently and calibrate multiple local metrics

Fig. 5 Two test images (*first column*) from ImageCLEF-11 (*top panel*) and NUS-WIDE (*bottom panel*) with their ten nearest neighbors selected using a global metric and local metrics. Images enclosed in blue rectangles are considered to be irrelevant neighbors of the test images selected by the global metric. Specific labels of local metrics used to select neighbors are depicted. For the test image in ImageCLEF-11, we use the global metric of the GM-LMNN method and local metrics of the MTLM-LMNN method, and for the test image in NUS-WIDE, we used the global metric of GM-LDML and local metrics of MTLM-LDML

implicitly, the joint learning approach in the proposed methods is more efficient and avoids the potential risk of information loss in the extra calibration procedure. Furthermore, the shared metric $\boldsymbol{w}_*$ in (6) plays the role of the global metric in modeling the correlations of labels. Thus, the discrimination of each specific label and commonalities of all labels can both be incorporated in each label-specific local metric in the proposed methods.

The proposed methods using both the discriminant manner and large-margin manner have advantages on two datasets. On ImageCLEF-11 with a moderate number of images, the proposed MTLM-LMNN of large margin manner performs better than the MTLM-LDML, yielding a higher MAP and smaller K_{opt}. In contrast, on the larger dataset NUS-WIDE, MTLM-LDML using the discriminant manner outperforms MTLM-LMNN. We can explain this result since on the NUS-WIDE dataset, the number of image pairs in each label's semantic cluster is much larger, and thus for the large-margin learning manner, the pairwise constraints of pulling similar images closer and pushing dissimilar images further away are seriously strict in the high dimensional feature space, which is prone to overfitting. On the other hand, the discriminant learning manner, which identifies positive/negative binary responses of pairwise constraints, is more flexible and less restrictive when using the linear representation of the distance metric in a high dimensional feature space.

4.4.2 Prediction results for rare and frequent labels

In a further experiment, we examine the prediction results of each label in depth and discussed the potential capacity of the proposed methods to improve the prediction of *rare* labels. Here we consider labels that occur less frequently than the average occurrence of all labels as *rare* labels, and the others as *frequent* labels (see the last column in Table 1). For the NUS-WIDE dataset, all 81 labels were divided into 57 *rare* labels and 24 *frequent* labels. To make a comprehensive comparison, we select two typical methods from the baselines, GM-LDML using global metric learning and LSLM-LDML using independent label-specific local metrics learning, as well as the two proposed methods: MTLM-LDML and MTLM-LMNN using multi-task local metrics learning.

Figure 6 shows the average precision (AP) of each label on NUS-WIDE obtained by the baselines and proposed methods. In Fig. 6a, the AP of each *rare* label obtained by the

proposed method MTLM-LDML is noticeably better than the results of both the global metric learning method GM-LDML and the independent label-specific local metric learning method LSLM-LDML. In particular, the mean AP results of all 57 *rare* labels obtained by GM-LDML, LSLM-LDML, and MTLM-LDML are 0.1173, 0.1421, and 0.159, respectively, realizing an improvement of 11.9% (comparing the results of LSLM-LDML and MTLM-LDML). In Fig. 6b, the mean AP results of all 24 *frequent* labels obtained by GM-LDML, LSLM-LDML, and MTLM-LDML are 0.2879, 0.3576, and 0.3709, respectively, realizing an improvement of 3.72% (comparing the results of LSLM-LDML and MTLM-LDML). Here we address the advantage of the proposed methods to improve the precision of *rare* labels. On one hand, using the local metrics of labels is more efficient for capturing the discriminations of the *rare* labels, which is significant in predicting these *rare* labels. On the other hand, the MTL framework in the proposed methods is able to model the correlations of labels, which is beneficial in bridging the associations between *rare* labels and *frequent* labels based on their co-occurrences.

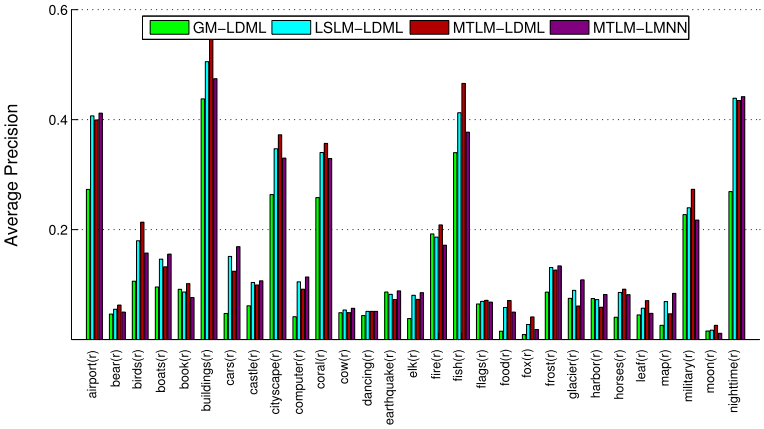
4.5 Overall annotation performance

Moreover, we summarize the best label prediction results of the proposed methods and baselines on the two datasets. We also include the state-of-the-art (S-O-T-A) results from other remarkable works for a comprehensive evaluation.

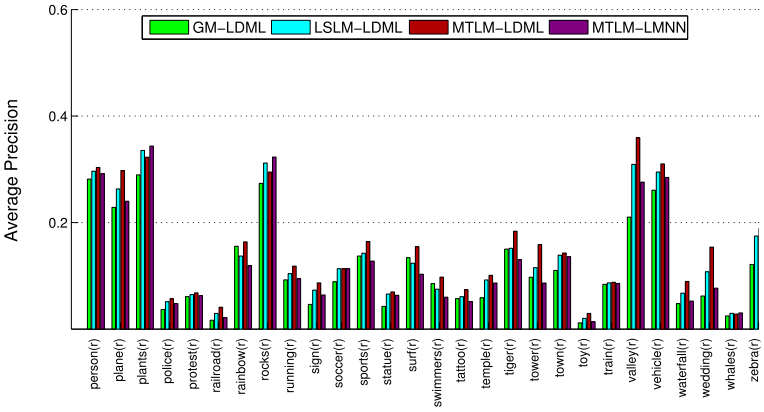
Table 2 gives an overview of the annotation performance in terms of MAP and F-ex[2] of the proposed methods, baselines, and S-O-T-A methods on the ImageCLEF-11 dataset. The S-O-T-A methods include those that achieved the top three results in the ImageCLEF2011 photo annotation contest. From Table 2 we can observe that: (1) Our proposed methods significantly improve both measures compared with the baselines, which confirms the advantage of using local metrics for label prediction and the superiority of leveraging the MTL framework to learn multiple local metrics simultaneously. (2) The performance of the proposed method MTLM-LMNN is competitive when compared with the S-T-O-A methods in the contest. CAEN [28] and ISIS [30], which are both SVM based methods, learn an SVM classifier for each label using sophisticated features (BoW histograms and Fisher vectors), while TUBFI also utilizes an MTL framework to learn from the integrated SVM outputs of all the labels. Here we address the importance and necessity of the proposed methods, which explore more efficient NN based methods combined with metric learning thereby providing a novel perspective for image annotation tasks.

Table 3 shows the overall annotation performance in terms of MAP measure for the NUS-WIDE dataset. As can be seen, our proposed methods consistently perform better than the baselines. In particular, the proposed MTLM-LDML achieves state-of-the-art performance with a MAP of 0.207, which is superior to the result of LSMP [8] with a MAP of around 0.193. The reasons for the improvement are two respects. First, more informative high dimensional multiple feature descriptors are utilized in the proposed method. Secondly, the proposed method can select a more appropriate neighborhood for the test image from the fully labeled training data using supervised learning approach, whereas LSMP uses a semi-supervised approach, which can result in noisy samples when constructing the similarity graph for both labeled and unlabeled images.

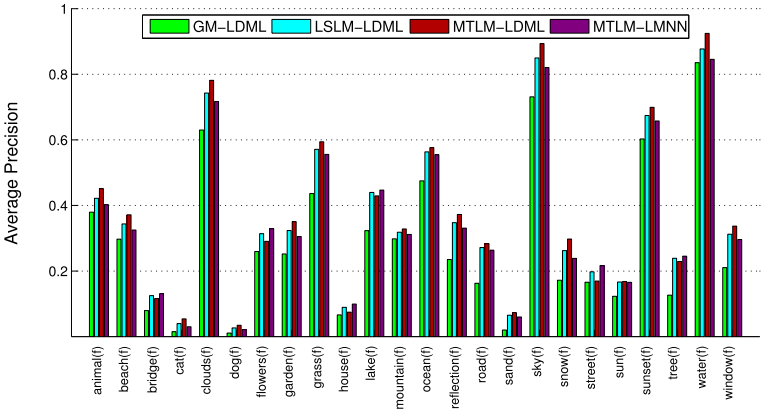
[2]The F-ex metric is an example-based evaluation, which is the averaged F1 score ($F1 = 2\frac{Precision*Recall}{Precision+Recall}$) of all images. Note that higher F-ex score implies better performance



(a) AP for *rare* labels: from "airport" to "nighttime".



(b) AP for *rare* labels: from "person" to "zebra".



(c) AP for *frequent* labels: from "animal" to "window".

Fig. 6 Average precisions for *rare* labels (**a**) and (**b**), denoted as "*label(r)*", and *frequent* labels (**c**), denoted as "*label(f)*", on NUS-WIDE dataset

Table 2 Annotation performance in terms of MAP and F-ex of the proposed methods, baselines and S-O-T-A methods on the ImageCLEF dataset

	Method	MAP	F-ex
Baselines	JEC [22]	0.166	0.298
	Tagprop [14]	0.247	0.423
	GM-LDML [15]	0.239	0.409
	GM-LMNN [32]	0.263	0.431
	LSLM-LDML [37]	0.299	0.457
	LSLM-LMNN	0.317	0.495
	SVM	0.325	0.521
Proposed	MTLM-LDML	0.356	0.573
	MTLM-LMNN	0.378	0.581
S-O-T-A	TUBFI [3]	0.388	0.552
	CAEN [28]	0.382	0.600
	ISIS [30]	0.375	0.612

Figure 7 shows some annotation results including successes and failures for proposed methods and baselines on both datasets, where for each sample the top five predicted labels are selected. We can observe from the successes that, proposed methods provide more accurate and related labels which are depicted in corresponding images. Especially on ImageCLEF-11, the proposed methods have the ability to boost abstract concepts, e.g., "*artificial*", "*cute*", "*inactive*", which further demonstrates the effectiveness of the proposed methods. However, the proposed methods may fail when the visual contents of image are highly variant to the ground-truth labels, e.g., "*melancholic*", "*book*", and "*ocean*" in two images in the last column of Fig. 7. It may lead proposed methods to select irrelevant neighbors for these images when formulating their semantic neighborhood, and further influence the label prediction results. Whereas in this condition, the general labels such as "*Neutral_Illumination*", "*No_Persons*", and "*water*" predicted by the baselines would be more appropriate to these images.

Table 3 Annotation performance in terms of MAP and F-ex of the proposed methods, baselines and S-O-T-A methods on NUS-WIDE dataset

	Method	MAP
Baselines	JEC [22]	0.129
	Tagprop [14]	0.146
	GM-LDML [15]	0.131
	GM-LMNN [32]	0.138
	LSLM-LDML [37]	0.187
	LSLM-LMNN	0.176
	SVM	0.173
Proposed	MTLM-LDML	0.207
	MTLM-LMNN	0.188
S-O-T-A	LSMP [8]	0.193

Image CLEF-11				
Ground truth	Neutral_Illumination, No_Blur, No_Persons	No_Persons, Partly_Blurred, Neutral_Illumination, inactive	Indoor, Partly_Blurred, Single_Person, natural, calm	Neutral_Illumination, No_Blur, No_Persons, melancholic, inactive
Baseline	**LSLM-LMNN:** Neutral_Illumination, Visual_Arts, No_Blur, No_Persons, inactive	**Tagprop:** Neutral_Illumination, Still_Life, Macro, Visual_Arts, boring	**GM-LMNN:** Family_Friends, Out_of_focus, Indoor, Partly_Blurred, Neutral_Illumination	**JEC:** Neutral_Illumination, Partly_Blurred, No_Persons, inactive, natural
Proposed	**MTLM-LMNN:** Neutral_Illumination, No_Persons, No_Blur, happy, cute	**MTLM-LMNN:** Neutral_Illumination, No_Persons, Partly_Blurred, Still_Life, inactive	**MTLM-LMNN:** Single_Person, Partly_Blurred, Day, bodypart, Indoor	**MTLM-LMNN:** Indoor, funny, Neutral_Illumination, Visual_Arts, Aesthetic_Impression
NUS-WIDE				
Ground truth	glacier, mountain, sky	clouds, moon, reflection, sky	car, clouds, fire, house	book, ocean, water
Baseline	**JEC:** mountain, castle, sky, sunset, reflection	**LSLM-LDML:** reflection, sky, mountain, sunset, sand	**GM-LDML:** building, car, field, shirt, street	**Tagprop:** sky, clouds, snow, water, reflection
Proposed	**MTLM-LDML:** mountain, sky, glacier, snow, water	**MTLM-LDML:** moon, sky, reflection, cityscape, buildings	**MTLM-LDML:** car, road, fire, building, house	**MTLM-LDML:** police, protest, window, street military

Fig. 7 Example annotation results including successes (*left* three samples on each panel) and failures (*right* one sample on each panel) for the proposed two methods and baselines on ImageCLEF-11 (*top panel*) and NUS-WIDE (*bottom panel*) datasets

4.6 Discussion

Finally, we give in-depth analysis on proposed methods and discuss practical issues such as weak labelling of dataset and model efficiency.

Weak labelling The weak labelling issue indicates that not all the objects of an image are explicitly labeled by human annotators while building an annotation dataset [32]. Indeed, the two datasets used in our work suffer from weak labelling. As shown in Table 1, the average number of labels per image of NUS-WIDE dataset is 1.76, which is generally not enough to depict all the objects of an image. Similarly, the situation of ImageCLEF-11 dataset is discussed in [3].

Here we would like to point out that our proposed methods are advantageous in tackling the weak labelling issue. First, when selecting the neighbor images for a test image, if we obtain nearest neighbors from the entire training data (as the NN based methods of [14, 22, 32]) using a global metric, then the missing labels can hardly appear among these neighbors hence make no contribution in the label prediction procedure. In contrary, our proposed methods formulate the neighborhood of a test image by choosing images from

Table 4 Running time (unit: hours) comparison of different distance metric learning methods for image annotation task on NUS-WIDE dataset. Here the label prediction time represents prediction time of all the test images

	Method	Metric learning time	Label prediction time
Baselines	JEC [22]	-	0.2
	Tagprop [14]	13.5	0.3
	GM-LDML [15]	12.2	0.4
	GM-LMNN [32]	15.8	0.4
	LSLM-LDML [37]	10.4	0.3
	LSLM-LMNN	14.7	0.2
Proposed	MTLM-LDML	20.3	0.3
	MTLM-LMNN	31.9	0.2

each label's semantic cluster (similar as in [32, 37]) using corresponding local metric, where all the labels are present explicitly. Thus, the missing labels have better chances of being assigned to a test image when predicting them with corresponding local metrics. Second, the proposed methods provide a multi-task framework for learning local metric for each label in a joint manner, which can incorporate inter-label relationships efficiently. However, previous discriminative methods [5, 37] that learn a classifier or local metric for each label in an independent manner generally ignore the inter-label relations. Although incremental calibration procedure can be used for compensation, it may still result in information loss.

In Fig. 7, several exemplars are weakly labeled images as the number of ground-truth labels is less than the number of our predictions. For example, it can be seen that for the two images in the second column, the proposed methods predict additional labels such as "*Still_Life*", "*cityscape*", "*building*". These labels are actually depicted in the corresponding images , but they are missing in ground-truth annotations.

Model efficiency The computational complexity of training our proposed methods generally consists of two parts: the cost of neighborhood selection and the cost of learning multiple local metrics. For selecting neighborhood of a training image, our proposed methods have the same complexity as previous NN based methods [14, 22, 32, 37]. In practice, rather than using the naive exhaustive comparison, the local sensitive hashing (LSH) based method [16] can be further utilized to accelerate the neighborhood selection in high dimensional feature space.

In Table 4, we provide the recorded running time for different methods on NUS-WIDE dataset. Generally, the running time consists of metric learning time (for training) and label prediction time (for testing).[3] We can make four observations: (1) The baselines that learn label-specific local metrics independently (e.g., LSLM-LDML and LSLM-LMNN) require less training time than baselines that learn a global metric (e.g., GM-LDML and GM-LMNN), since learning local metrics is performed on subsets (semantic clusters) rather than the entire training set. Note that, JEC does not require training as it uses the simple average of base distance metrics to define neighbors. (2) To learn multiple local metrics, the

[3]Here the time of neighborhood selection is not included in recording the running time. In our experiment, exhaustive neighborhood selection for all the training and test images takes around 24 and 18 hours respectively

proposed methods generally have higher computational complexity than the baselines. The reason is that the proposed methods need to learn label-specific local metrics and a shared metric jointly, as described in Algorithm 1 in Appendix A. However, since the calibration step for local metrics used in [37] is not necessary in multi-task learning framework, the computational complexity of proposed methods can be efficiently reduced. Besides, in our experiment, we have found that a potential way to boost the training efficiency of the proposed methods is to leverage the pre-trained global metric from GM-LDML and GM-LMNN, or label-specific local metrics from LSLM-LDML and LSLM-LMNN as hot starting point for the proposed methods. This strategy can facilitate the convergence of optimization procedure for proposed methods. (3) For the proposed methods, the training time of MTLM-LMNN is less than that of MTLM-LDML, because MTLM-LMNN requires additional computation to assess the "active" pairwise constraints in each iteration as shown in Appendix B. (4) Once we have learnt local metrics for proposed methods, the label prediction procedure for a test image takes similar time as the baselines (see the label prediction schemes in (2) and (4). Note that, in practice, the label prediction time of the proposed methods and baselines might be slightly variant, as the optimal number of nearest neighbors required in these methods is different (see Section 4.4.1).

5 Conclusion

In this paper, we revisited existing metric learning works focusing on image annotation problems. We found that these works largely aim to learn a unified global metric to measure the image-level similarity of image pairs. This may lead to potential limitations in modeling the discrimination of different labels. In addition, to extend our previous work which learns multiple label-specific local metrics independently, we leveraged MTL framework to learn multiple local metrics jointly, and introduced an additional shared metric to model the correlations between labels explicitly. We proposed two local metric learning methods based on discriminant manner and large-margin manner, respectively, in the MTL framework. Both methods have their own advantages depending on practical settings. We also developed efficient optimization algorithms for the two methods. Extensive experiments on two challenging datasets demonstrate the superiority of the proposed methods compared with previous global metric learning methods and independent local metric learning methods.

In the future, we will investigate the scalability of the proposed methods with respect to the number of labels, since under web-based conditions, labels are free and large scale, and thus it is more reasonable to extend one local metric to a cluster of related labels rather than a single label. We will also intend extending the current MTL framework to multiple regularization terms. Currently only one shared metric is utilized to model the commonalities of all labels, but it is possible to introduce multiple shared metrics since labels are locally correlated.

Appendix A: Optimization in MTLM-LDML

For the logistic regression model in (7), the bias parameter b_m can be considered to be an additional entry in the weight vector $\boldsymbol{w}_m$, as $b_m = w_0$ and $d^0(\cdot) = -1$. Thus, the probability

of a similar pair in (7) can be concisely expressed as

$$p_{qp}(s_{qp} = 1|y_m; D_{y_m}(I_q, I_p), b) = \sigma(-D_{y_m}(I_q, I_p)). \tag{13}$$

Intuitively, to simplify the denotation, the cost function in (9) can be rewritten as

$$E(\boldsymbol{w}) = \gamma_* \|\boldsymbol{w}_*\|_2^2 + \sum_{m=1}^{M} \left[\gamma_m \|\boldsymbol{w}_m\|_2^2 + \sum_{(I_q, I_p) \in \mathcal{P}_{y_m}} \log(1 + \exp(s_{qp} D_{y_m}(I_q, I_p))) \right]. \tag{14}$$

Based on the definitions of D_{y_m} and the sigmoid function $\sigma(z) = \frac{1}{1+\exp(-z)}$, we first consider the gradients with respect to weight vectors $\{\boldsymbol{w}_m\}_{m=1}^M$ of the label-specific metrics as

$$E(\boldsymbol{w}_m) = \gamma_m \|\boldsymbol{w}_m\|_2^2 + \sum_{(I_q, I_p) \in \mathcal{P}_{y_m}} \log(1 + \exp(s_{qp} D_{y_m}(I_q, I_p))). \tag{15}$$

Hence, the gradient of $\boldsymbol{w}_m$ is given by

$$\boldsymbol{G}(\boldsymbol{w}_m) = \frac{\partial E(\boldsymbol{w})}{\partial \boldsymbol{w}_m} = 2\gamma_m \boldsymbol{w}_m + \sum_{(I_q, I_p) \in \mathcal{P}_{y_m}} s_{qp} \sigma(s_{qp} D_{y_m}(I_q, I_p)) \boldsymbol{d}(I_q, I_p). \tag{16}$$

Then, we consider gradients with respect to weight vector $\boldsymbol{w}_*$ of the shared metric as

$$\boldsymbol{G}(\boldsymbol{w}_*) = \frac{\partial E(\boldsymbol{w})}{\partial \boldsymbol{w}_*} = 2\gamma_* \boldsymbol{w}_* + \sum_{m=1}^{M} \sum_{(I_q, I_p) \in \mathcal{P}_{y_m}} s_{qp} \sigma(s_{qp} D_{y_m}(I_q, I_p)) \boldsymbol{d}(I_q, I_p). \tag{17}$$

Note that the gradients of $\{\boldsymbol{w}_m\}_{m=1}^M$ in (16) can be calculated separately according to the pairwise constraints of labels $\{y_m\}_{m=1}^M$, while the gradient of $\boldsymbol{w}_*$ in (17) depends on all gradients of $\{\boldsymbol{w}_m\}_{m=1}^M$ and must be calculated thereafter.

Finally, the iterative update solutions for weight vectors $\{\boldsymbol{w}_m\}_{m=1}^M$ and $\boldsymbol{w}_*$ can be formulated as

$$\boldsymbol{w}_m^{t+1} = [\boldsymbol{w}_m^t - \eta_t \boldsymbol{G}(\boldsymbol{w}_m^t)]_+, \tag{18}$$

$$\boldsymbol{w}_*^{t+1} = [\boldsymbol{w}_*^t - \eta_t \boldsymbol{G}(\boldsymbol{w}_*^t)]_+, \tag{19}$$

where η_t is the step size of iteration t, and $[f]_+ = \max(0, f)$ truncates any negative entries in $\boldsymbol{w}$ and sets them to zero for the non-negative constraints of $\boldsymbol{w}$. Algorithm 1 summarizes the learning process in the proposed MTLM-LDML method.

Algorithm 1 Gradient descent based optimization in MTLM-LDML

Input: A set of labeled training images $\mathcal{C} = \{\{I_1, Y_1\}, ..., \{I_D, Y_D\}\}$ with pairwise constraints $\mathcal{P}_{y_m}$ for each label.
Output: Weight vectors $\{\boldsymbol{w}_m\}_{m=1}^M$ of label-specific metrics and $\boldsymbol{w}_*$ of the shared metric.

1: Initialize $\boldsymbol{w}_* = \mathbf{1}$, $\{\boldsymbol{w}_{m=1}^M\} = \mathbf{1}$ for iteration $t = 0$, $\rho = \frac{\frac{1}{M}\sum_{m=1}^M \gamma_m}{\gamma_*} = 5$.
2: **repeat**
3: Set $t = t + 1$, step size $\eta_t = 1/t$.
4: Update $\{\boldsymbol{w}_m\}_{m=1}^M$ according to (18).
5: Update $\boldsymbol{w}_*$ according to (19).
6: **until** Cost function $E(\boldsymbol{w})$ reaches convergence in (14).

Appendix B: Optimization in MTLM-LMNN

Similar to the computation of gradients in Appendix A, from the cost function in (12) we can derive the gradients of weight vectors $\{\boldsymbol{w}_m\}_{m=1}^M$ and $\boldsymbol{w}_*$ as

$$\boldsymbol{G}(\boldsymbol{w}_m) = 2\gamma_m \boldsymbol{w}_m + \sum_{(I_q, I_n)\in\mathcal{P}_{y_m}^+} \boldsymbol{d}(I_q, I_p) + \mu \sum_{(I_q, I_n)\in\mathcal{P}_{y_m}^-} \big(\boldsymbol{d}(I_q, I_p) - \boldsymbol{d}(I_q, I_n)\big), \quad (20)$$

$$\boldsymbol{G}(\boldsymbol{w}_*) = 2\gamma_* \boldsymbol{w}_* + \sum_{m=1}^{M} \left[\sum_{(I_q, I_n)\in\mathcal{P}_{y_m}^+} \boldsymbol{d}(I_q, I_p) + \mu \sum_{(I_q, I_n)\in\mathcal{P}_{y_m}^-} \big(\boldsymbol{d}(I_q, I_p) - \boldsymbol{d}(I_q, I_n)\big) \right]. \quad (21)$$

Here we make the similar observations that the gradient of $\{\boldsymbol{w}_m\}_{m=1}^M$ in (20) can be calculated separately according to the pairwise constraints $\mathcal{P}_{y_m}$ of each label y_m, while the gradient of $\boldsymbol{w}_*$ in (21) depends on all the gradients of $\{\boldsymbol{w}_m\}_{m=1}^M$.

Since naive computation of the gradients in (20) and (21) would be extremely expensive, we consider only the "*active*" triplets in $\mathcal{P}_{y_m}$ that change from iteration t to iteration $t+1$ based on the following update rules:

$$\begin{aligned}\boldsymbol{G}(\boldsymbol{w}_m^{t+1}) = \boldsymbol{G}(\boldsymbol{w}_m^{t}) - \eta_t \sum_{(I_q, I_p, I_n)\in\mathcal{P}_{y_m}^t - P_{y_m}^{t+1}} (\boldsymbol{d}(I_q, I_p) - \boldsymbol{d}(I_q, I_n)) + \eta_t \\ \sum_{(I_q, I_p, I_n)\in\mathcal{P}_{y_m}^{t+1} - P_{y_m}^{t}} (\boldsymbol{d}(I_q, I_p) - \boldsymbol{d}(I_q, I_n)),\end{aligned} \quad (22)$$

$$\begin{aligned}\boldsymbol{G}(\boldsymbol{w}_*^{t+1}) = \boldsymbol{G}(\boldsymbol{w}_*^{t}) - \sum_{m=1}^{M} \Bigg[\eta_t \sum_{(I_q, I_p, I_n)\in\mathcal{P}_{y_m}^t - \mathcal{P}_{y_m}^{t+1}} (\boldsymbol{d}(I_q, I_p) - \boldsymbol{d}(I_q, I_n)) + \eta_t \\ \sum_{(I_q, I_p, I_n)\in\mathcal{P}_{y_m}^{t+1} - \mathcal{P}_{y_m}^{t}} (\boldsymbol{d}(I_q, I_p) - \boldsymbol{d}(I_q, I_n)) \Bigg],\end{aligned} \quad (23)$$

where μ_t is the step size in the t-th iteration, $\mathcal{P}_{y_m}^{t+1} - \mathcal{P}_{y_m}^{t}$ represents the new triplets appearing in $\mathcal{P}_{y_m}^{t+1}$, and $\mathcal{P}_{y_m}^{t} - \mathcal{P}_{y_m}^{t+1}$ represents the old triplets that have disappeared in $\mathcal{P}_{y_m}^{t+1}$. For a small step size, the set $\mathcal{P}_{y_m}^{t}$ changes little in each iteration. In this case, computing the gradients in (22) and (23) is much faster.

Finally, we utilize a gradient descent based learning process similar to that depicted in Algorithm 1 to learn weight vectors $\{\boldsymbol{w}_m\}_{m=1}^M$ and $\boldsymbol{w}_*$ for the proposed MTLM-LMNN model.

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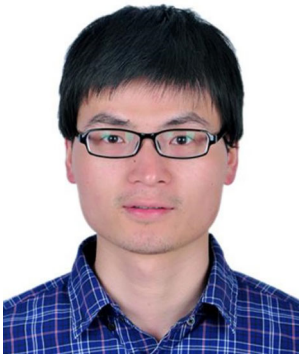

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